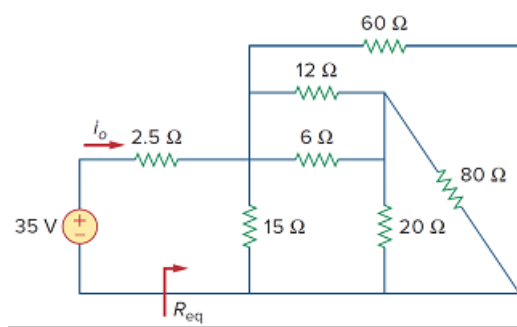


Problem

Find R_{eq} and i_o in the circuit of Fig. 2.102.

Figure 2.102:



Step-by-step solution

Step 1 of 7

Consider the following circuit diagram:

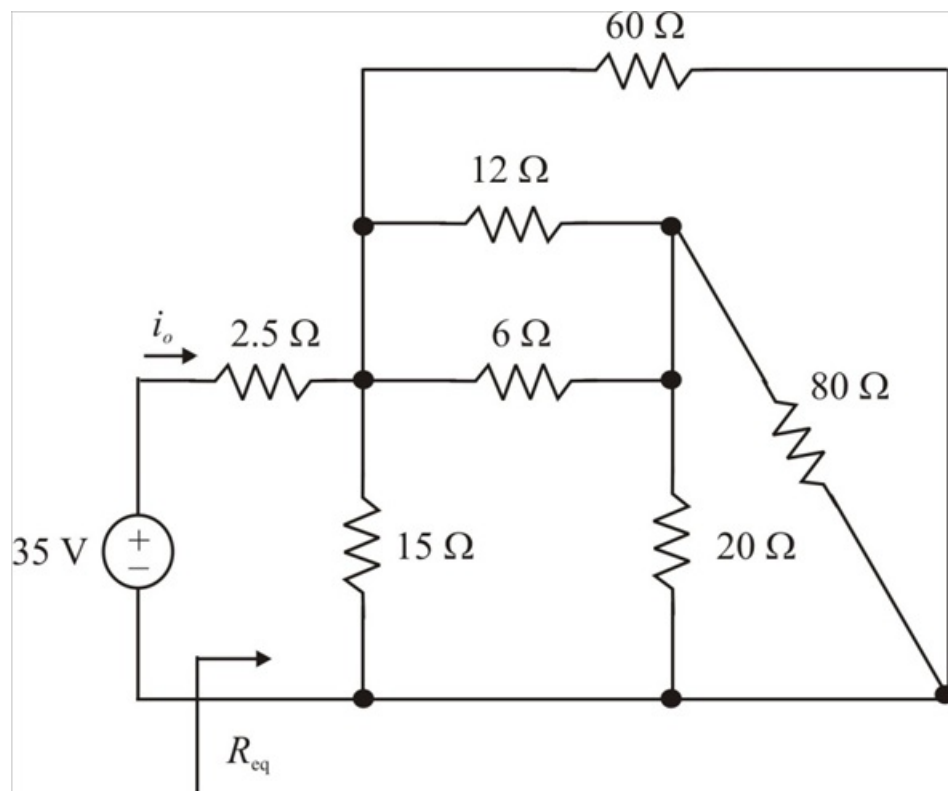


Figure 1

Step 2 of 7

To get the R_{eq} , we combine the resistors in series and in parallel.

The $12\ \Omega$ and $6\ \Omega$ resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 12\ \Omega \parallel 6\ \Omega &= \frac{12 \times 6}{12 + 6} \\ &= 4\ \Omega \end{aligned}$$

The $20\ \Omega$ and $80\ \Omega$ resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 20\ \Omega \parallel 80\ \Omega &= \frac{20 \times 80}{20 + 80} \\ &= 16\ \Omega \end{aligned}$$

Step 3 of 7

Then the reduced circuit is shown in Figure 2.

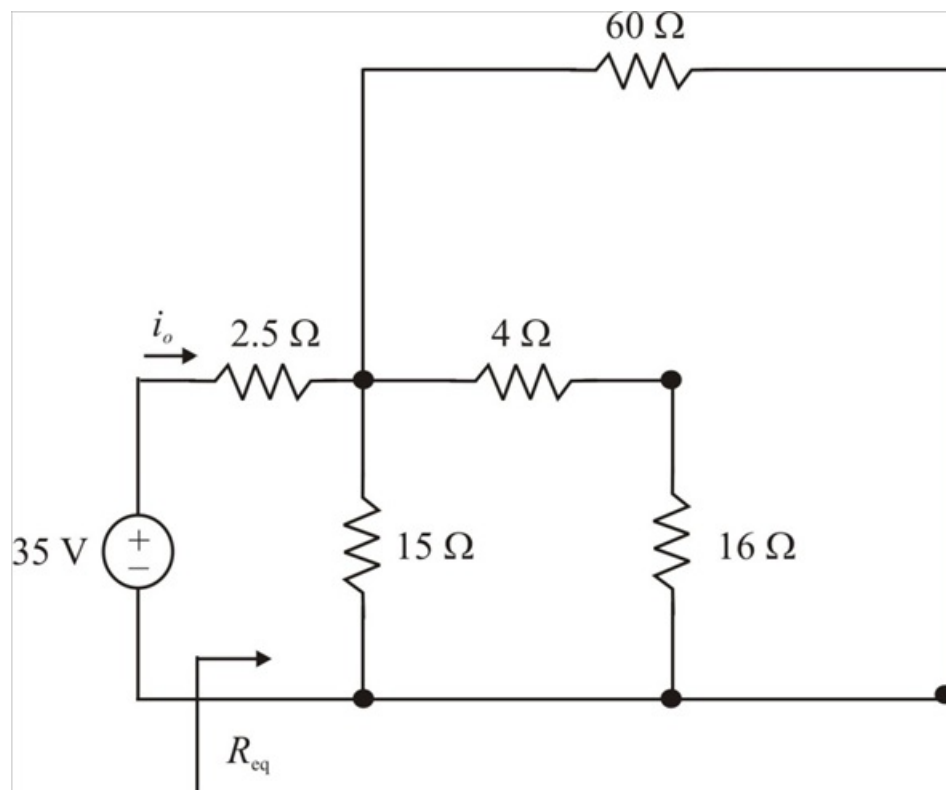


Figure 2

Step 4 of 7

The $4\ \Omega$ and $16\ \Omega$ resistors are in series. Their combined resistance is,

$$4\ \Omega + 16\ \Omega = 20\ \Omega$$

This $20\ \Omega$ and $60\ \Omega$ resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 20\ \Omega \parallel 60\ \Omega &= \frac{20 \times 60}{20 + 60} \\ &= 15\ \Omega \end{aligned}$$

Step 5 of 7

Then the reduced circuit is shown in Figure 3.

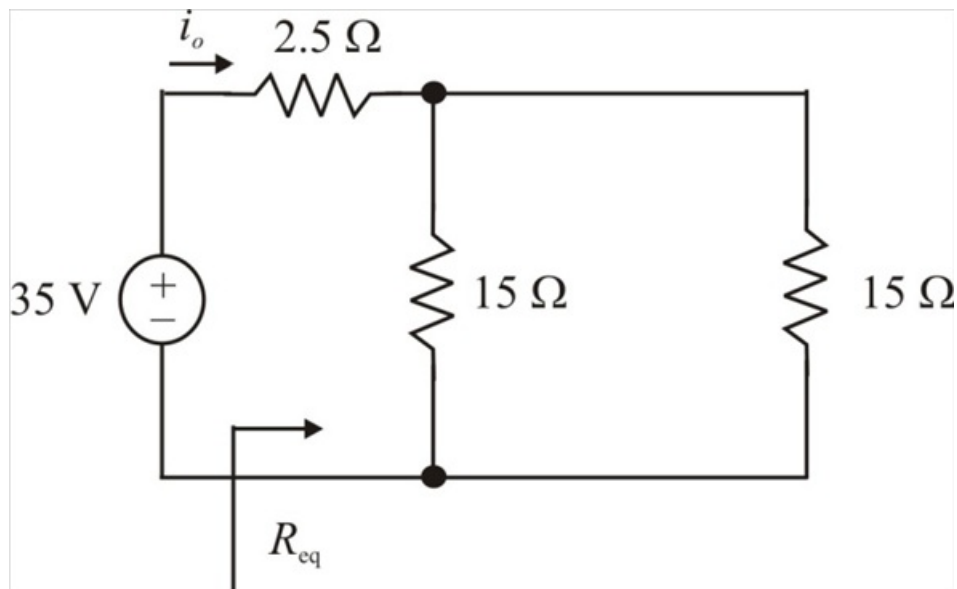


Figure 3

Step 6 of 7

The 15 Ω and 15 Ω resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 15\Omega \parallel 15\Omega &= \frac{15 \times 15}{15 + 15} \\ &= 7.5\Omega \end{aligned}$$

This 7.5 Ω and 2.5 Ω resistors are in series. Their combined resistance is,

$$\begin{aligned} R_{eq} &= 7.5\Omega + 2.5\Omega \\ &= 10\Omega \end{aligned}$$

Therefore the equivalent resistance R_{eq} seen by the source is $\boxed{10\Omega}$.

Step 7 of 7

By using Ohms law, the current i_o can be calculated as follows.

$$\begin{aligned} i_o &= \frac{v}{R} \\ &= \frac{35}{10} \\ &= 3.5\text{ A} \end{aligned}$$

Therefore, the value of current i_o is $\boxed{3.5\text{ A}}$.